

CampusWatt:
An AI-Powered Energy Forecasting and
Decision Support Platform for Campus Buildings

PHASE I REPORT

Submitted by

AAKASH V	230701002
ADITHYA J	230701013
AKSHAYA BALAJI N	230701023
ADHAVAN BALAJI N M	230701012

*in partial fulfillment for the award of the
degree*

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE,
CHENNAI ANNA UNIVERSITY:CHENNAI-025

September 2026

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Report titled “**CAMPUSWATT: AN AI-POWERED ENERGY FORECASTING AND DECISION SUPPORT PLATFORM FOR CAMPUS BUILDINGS**” is the bonafide work of **AAKASH V (230701002), ADITHYA J (230701013), AKSHAYA BALAJI N (230701023), ADHAVAN BALAJI N M (230701012)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

This project addresses the following Sustainable Development Goals:

SDG 7 - Affordable and Clean Energy: Improves energy efficiency in institutional buildings through accurate demand forecasting and the identification of avoidable consumption.

SDG 9 - Industry, Innovation and Infrastructure: Applies artificial intelligence to build intelligent, resilient energy management infrastructure for educational institutions.

SDG 12 - Responsible Consumption and Production: Enables data-driven reduction of wasteful electricity consumption through anomaly detection and cost savings recommendations.

SIGNATURE

Dr. J. Manoranjini, Ph.D.,

Professor and Head,
Department of Computer Science
and Engineering,
Rajalakshmi Engineering College,
Chennai – 602 105.

SIGNATURE

Dr. V.Prakash

Associate Professor
Department of Computer Science
and Engineering,
Rajalakshmi Engineering College,
Chennai – 602 105.

Submitted to Project Viva-Voce Examination held on _____

Internal Examiner

External Examiner

ABSTRACT

Educational institutions consume a significant share of their operating budget on electricity, yet campus energy is typically managed with a single monthly bill and no visibility into where, when or why consumption occurs. Without accurate demand forecasting, faults and wasteful usage remain undetected for weeks, peak demand breaches attract avoidable tariff penalties, and administrators have no quantified basis on which to approve efficiency measures.

CampusWatt: An AI-Powered Energy Forecasting and Decision Support Platform is proposed to address this gap for a single academic block. The system constructs an hourly consumption dataset from an audited room-by-room equipment inventory, combined with local weather patterns and the academic calendar, producing 43,800 records across five functional zones for a full year.

A gradient-boosting regression model is trained per zone using lag features, rolling statistics, cyclical time encodings, weather variables and academic day-type indicators. On a held-out test period the model attains a block-level Mean Absolute Percentage Error of 5.75 percent, a Mean Absolute Error of 0.96 kWh per hour and a coefficient of determination of 0.987, improving on a seasonal-naive baseline by between 80 and 98 percent depending on zone.

Anomalies are detected as significant deviations between measured consumption and the model forecast rather than against the previous day, which would misfire on every regime change. Detected excess load is then attributed to specific equipment through a signature matching procedure that divides the excess power by the rated wattage of each installed device type and eliminates explanations requiring more units than physically exist. This converts detection into diagnosis.

The platform further models the complete High Tension electricity tariff, including energy charge, maximum demand charge, power factor penalty and statutory levies. Analysis reveals that the demand charge exceeds the energy charge, establishing that peak management yields greater financial benefit than proportional consumption reduction. A what-if simulation engine and a ranked recommendation module quantify savings, capital cost and payback period for each candidate measure, identifying Rs 2,37,264 of annual savings of which the majority requires no capital expenditure.

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S. MEGANATHAN, B.E., F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S. N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J. MANORANJINI, Ph.D.**, Professor and Head of the Department of Computer Science and Engineering for her guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Dr.V.Prakash**, Department of Computer Science and Engineering, for the valuable guidance throughout the course of the project.

AAKASH V	230701002
ADITHYA J	230701013
AKSHAYA BALAJI N	230701023
ADHAVAN BALAJI N M	230701012

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	4
	ACKNOWLEDGEMENT	6
	LIST OF TABLES	8
	LIST OF FIGURES	9
	LIST OF ABBREVIATIONS	10
1.	INTRODUCTION	11
	1.1 GENERAL	11
	1.2 OBJECTIVE	12
	1.3 EXISTING SYSTEMS	13
	1.4 PROPOSED SYSTEM	14
2.	LITERATURE SURVEY	16
	2.1 LITERATURE REVIEW	16
	2.2 COMPARISON AND DISCUSSION	18
	2.3 CONCLUSION	19
3.	SYSTEM DESIGN	20
	3.1 SYSTEM ARCHITECTURE	20
	3.2 SYSTEM REQUIREMENTS	21
	3.2.1 Software Requirements	21
	3.2.2 Hardware Requirements	22
	3.2.3 Dataset Requirements	23
	3.2.4 Deployment and Scaling	24
	3.3 SYSTEM DESIGN	24
	3.3.1 Activity Diagram	24

	3.3.2 Data Flow Diagram	25
4.	PROJECT DESCRIPTION	27
	4.1 METHODOLOGIES	27
	4.2 MODULES	27
	4.2.1 Dataset Generation and Preprocessing	27
	4.2.2 Feature Engineering	28
	4.2.3 Model Training and Evaluation	29
	4.2.4 Anomaly Detection and Signature Matching	30
	4.2.5 Tariff Modelling and Explainability	31
	4.2.6 Simulation, Recommendation and Presentation	32
5.	IMPLEMENTATION AND RESULT DISCUSSION	34
	5.1 IMPLEMENTATION	34
	5.2 DATASET CHARACTERISTICS	34
	5.3 FORECASTING RESULTS	35
	5.4 ANOMALY DETECTION AND ATTRIBUTION	36
	5.5 FINANCIAL ANALYSIS AND RECOMMENDATIONS	38
	5.6 DISCUSSION	40
6.	CONCLUSION AND FUTURE WORK	42
	6.1 CONCLUSION	42
	6.2 FUTURE WORK	42
	REFERENCES	44

LIST OF FIGURES

FIGURE NO	FIGURE NAME
3.1	Architecture diagram of the proposed system
3.2	Activity diagram for the operational environment
3.3	Activity diagram for the maintenance environment
3.4	Data flow diagram of the system
5.1	Zone-wise share of block energy consumption
5.2	Block-level actual versus forecast consumption
5.3	Forecast error against seasonal-naive baseline, by zone
5.4	Composition of the monthly HT electricity bill
5.5	Monthly recorded peak demand against sanctioned limit

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CDD	Cooling Degree Day
CRUD	Create, Read, Update, Delete
HT	High Tension
HVAC	Heating, Ventilation and Air Conditioning
JWT	JSON Web Token
kVA	Kilovolt-Ampere
kWh	Kilowatt-Hour
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MD	Maximum Demand
ML	Machine Learning
NILM	Non-Intrusive Load Monitoring
NLU	Natural Language Understanding
PF	Power Factor
RBAC	Role-Based Access Control
REST	Representational State Transfer
SHAP	SHapley Additive exPlanations
TANGEDCO	Tamil Nadu Generation and Distribution Corporation
XAI	Explainable Artificial Intelligence

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Electricity is among the largest recurring operational expenses incurred by educational institutions. A single academic block containing classrooms, computer laboratories, staff offices and shared service areas can consume in excess of one hundred thousand units of electricity annually, representing a substantial and continuing financial commitment. Despite this scale of expenditure, energy in most Indian campuses is managed retrospectively through a single consolidated monthly bill.

This arrangement provides a total figure but no structure. Facilities staff can observe that consumption has risen, but cannot determine which zone was responsible, at what hour the increase occurred, whether it was attributable to weather, or whether it represented legitimate academic activity as opposed to equipment inadvertently left running. Corrective action, when taken at all, therefore occurs weeks after the underlying event and without evidence to direct it.

Three consequences follow. First, avoidable consumption persists undetected. Laboratory computers left powered overnight, air conditioning units operating in unoccupied offices, and corridor lighting running through vacation periods produce a continuous baseline load that is invisible in a monthly total. Second, gradual equipment degradation escapes notice entirely; a failing compressor that raises consumption by two percent each week is imperceptible in day-to-day comparison, yet costs more over a billing period than any single incident. Third, and most significantly in financial terms, High Tension consumers are billed not only for the energy they consume but additionally on the highest demand recorded during the billing period, meaning a brief coincident peak carries cost for the entire month.

Recent developments in machine learning provide practical means of addressing these deficiencies. Supervised regression models can learn the relationship between building consumption and its physical and scheduling drivers with sufficient accuracy that departures from expected behaviour become statistically detectable. Explainability techniques permit the resulting predictions to be decomposed into contributing factors

rather than presented as opaque outputs. Together these capabilities support a shift from retrospective billing analysis toward continuous, forward-looking energy management.

1.2 OBJECTIVE

The principal aim of CampusWatt is to convert raw electrical consumption data for a single academic block into specific, costed and actionable decisions for campus administrators. The system is designed to replace retrospective monthly reporting with continuous forecasting, automated fault identification and quantified recommendations.

The first objective is to forecast consumption at the level of individual functional zones rather than for the building as a whole, since a block-level total conceals the zone-specific behaviour on which any corrective action must be based. Forecasts are required to account for outside weather conditions and for the academic calendar, as an examination-period weekday exhibits a materially different consumption profile from an ordinary teaching day.

The second objective is to detect anomalous consumption by comparison against the model forecast rather than against historical readings, and thereafter to attribute the detected excess to specific categories of installed equipment. The intention is to produce findings that a maintenance technician can act upon directly, rather than alerts that merely report that consumption is abnormal.

The third objective is to model the complete electricity tariff applicable to High Tension institutional consumers, encompassing the energy charge, the maximum demand charge, the power factor penalty and statutory levies, so that every finding and recommendation may be expressed in monetary terms.

The fourth objective is to provide decision support in the form of a what-if simulation facility and a ranked recommendation engine in which each proposed measure carries its estimated annual saving, capital requirement, payback period and supporting evidence. The final objective is to present these capabilities through a role-based web interface accompanied by a conversational assistant capable of answering natural language queries grounded strictly in retrieved system data.

1.3 EXISTING SYSTEMS

Several categories of system currently address building energy management, each with limitations when applied to the specific circumstances of an academic block within an Indian engineering college.

Commercial Building Management Systems supplied by vendors such as Schneider Electric, Siemens and Honeywell provide comprehensive monitoring, automated control of HVAC plant and detailed reporting. These platforms are technically capable but presuppose extensive sub-metering infrastructure and dedicated control hardware. Their licensing and installation costs are calibrated for commercial office towers and industrial facilities, placing them beyond the practical reach of an individual academic department. They also assume the presence of trained facilities personnel to operate them.

Utility-provided smart meter portals, offered by distribution companies to High Tension consumers, present consumption and demand data at the level of the service connection. Such portals record what was consumed and confirm whether the sanctioned demand was exceeded, but perform no forecasting, no anomaly detection and no attribution. They are reporting instruments rather than decision support systems, and provide no visibility below the level of the whole connection.

General-purpose Internet of Things energy monitoring products, including the Sense home energy monitor and various commercial sub-metering solutions, offer appliance-level disaggregation for residential and small commercial premises. Their underlying load disaggregation techniques are designed for environments containing a limited number of distinct appliances with characteristic switching signatures. An academic block containing two hundred and forty identical desktop computers and two hundred and twenty-four identical ceiling fans presents a fundamentally different disaggregation problem, since the individual signatures are neither distinct nor separable.

Within the academic literature, a substantial body of work addresses building energy forecasting, and a further body addresses anomaly detection in consumption data. These contributions are methodologically valuable but are typically evaluated as isolated components on public benchmark datasets. They do not generally extend to attribution of detected anomalies to physical equipment, to modelling of the tariff structure under which the building is actually billed, or to the generation of recommendations expressed in the financial terms required for institutional approval.

A common deficiency runs through all four categories. Each addresses measurement, or prediction, or detection, but none closes the loop from measurement through to a costed decision that a campus administrator can act upon.

1.4 PROPOSED SYSTEM

The proposed CampusWatt architecture is organised as a layered pipeline in which each stage consumes the output of the preceding stage, progressing from raw consumption data to actionable recommendations.

The Data Acquisition and Preprocessing layer constructs the hourly consumption dataset. Rather than adopting a generic public dataset, consumption is synthesised bottom-up from an audited equipment inventory of the block, in which the rated wattage and installed quantity of every fan, light, computer, display panel, air conditioner and server is combined with realistic occupancy schedules, local weather variation and the academic calendar. This approach yields a dataset in which anomalous events are labelled by construction, permitting detection performance to be measured quantitatively rather than merely demonstrated.

The Forecasting layer trains one gradient-boosting regression model per functional zone. The feature set comprises lag values at one, twenty-four and one hundred and sixty-eight hours, rolling means and standard deviations, cyclical encodings of hour and weekday, outside temperature and humidity, derived cooling degree values, and binary indicators for working day, examination period, vacation and institutional event. A learned profile of mean consumption for each hour and day-type combination is included as an explicit feature, which permits the model to represent examination-period behaviour correctly even when every lagged value originates from an ordinary working day.

The Detection layer computes the residual between measured and forecast consumption and flags statistically significant departures. Two complementary detectors operate in parallel: a residual-based detector identifying acute events, and a rolling-baseline monitor identifying gradual upward drift characteristic of equipment degradation. Detected excess is subsequently passed to the signature matching procedure for attribution to specific equipment categories.

The Financial layer implements the complete High Tension tariff. Energy charge is computed with time-of-day banding, demand charge on the monthly maximum

recorded demand, and power factor penalty against the prescribed target, with statutory electricity duty and fixed charges applied. This layer converts every physical quantity produced by the preceding layers into monetary terms.

The Decision Support layer comprises the explainability module, which decomposes consumption change into weather, calendar, occupancy and residual components; the what-if simulator, which models arbitrary combinations of efficiency measures; and the recommendation engine, which ranks measures by annual saving and tiers them by capital requirement. The Presentation layer delivers these outputs through a role-based web application together with a conversational assistant whose responses are constrained to retrieved system values.

CHAPTER 2

LITERATURE SURVEY

This chapter reviews prior research relevant to the objectives of the proposed system, spanning building energy prediction, anomaly detection in consumption data, load disaggregation, model explainability and campus-scale energy management. The review establishes both the methodological foundations adopted by CampusWatt and the specific gap it addresses.

2.1 LITERATURE REVIEW

Amasyali and El-Gohary conducted a comprehensive review of data-driven building energy consumption prediction studies, surveying the modelling techniques, input features and evaluation practices reported across a large body of published work. Their analysis establishes that ensemble and tree-based methods consistently perform competitively against neural approaches on building consumption data, and that the selection of temporal and weather features frequently exerts greater influence on accuracy than the choice of learning algorithm. This finding directly informs the present work, in which feature engineering rather than model complexity was prioritised.

Chen and Guestrin introduced XGBoost, a scalable gradient boosting system that has since become a standard method for structured regression problems. The formulation addresses sparsity, provides regularisation to control overfitting and supports efficient training on moderate datasets. Gradient boosting of this family is adopted in the present work in preference to recurrent neural architectures, on the grounds that building consumption data is tabular, exhibits strong calendar structure, and is available in quantities better suited to boosted trees than to deep sequence models.

Ke and colleagues proposed LightGBM, introducing histogram-based split finding and leaf-wise tree growth to reduce training time substantially without loss of accuracy. The histogram-based gradient boosting implementation employed in the present work derives from this line of development, and permits the five zone-level models to be trained in seconds, making frequent retraining practical as new consumption data accumulates.

Seem examined the use of intelligent data analysis to detect abnormal energy consumption in buildings, proposing statistical methods for identifying readings that

depart from established patterns. The work establishes the principle that abnormality must be assessed relative to an expected value conditioned on operating context rather than against a fixed threshold. The present system extends this principle by conditioning the expectation on a learned forecast incorporating weather and academic calendar effects.

Himeur and colleagues reviewed artificial intelligence based anomaly detection of energy consumption in buildings, categorising approaches by learning paradigm and by the nature of the anomalies targeted. Their survey identifies the scarcity of labelled anomaly data as a persistent obstacle to rigorous evaluation, since most public datasets provide consumption readings without ground truth annotation of faults. The present work addresses this limitation directly by generating its dataset with anomalies injected under known parameters, thereby enabling detection performance to be measured against ground truth.

Hart established the field of non-intrusive load monitoring, demonstrating that individual appliance operation can be inferred from aggregate electrical measurements at a single point by analysing the step changes associated with switching events. This foundational work motivates the attribution approach adopted here. However, classical non-intrusive load monitoring assumes appliances with distinguishable signatures, an assumption that fails in an academic block containing hundreds of identical fans and computers. The present work therefore substitutes an inventory-constrained arithmetic attribution, in which candidate explanations requiring more units than are physically installed are eliminated.

Lundberg and Lee proposed SHapley Additive exPlanations, a unified framework for interpreting the predictions of machine learning models by assigning each input feature an additive contribution derived from cooperative game theory. The method provides both local explanations for individual predictions and aggregate feature importance. The explainability module of the present system applies this principle, though contributions are reported in monetary rather than dimensionless units, on the reasoning that administrative decisions are made in financial terms.

Miller and colleagues released the Building Data Genome Project 2, an open dataset comprising hourly electrical meter readings from over one thousand non-residential buildings, including a substantial number of educational facilities, accompanied by weather data and building metadata. This dataset was evaluated as a

candidate data source for the present work. It was ultimately not adopted, because its buildings are located predominantly in North American and European climates and their metadata does not extend to equipment inventories, which the attribution procedure requires.

Fan, Xiao and Wang applied data mining techniques to building operational performance data, demonstrating that patterns extracted from routinely collected building management data can identify both inefficiency and opportunity for improvement. Their work supports the position that value in building energy data arises less from measurement precision than from the analytical layer constructed above it, a position central to the design of the present system.

Considerable published work addresses demand-side management and peak load reduction in institutional settings, establishing that load shifting and scheduling adjustments can reduce maximum demand without reducing total energy consumed. This literature is directly relevant, since maximum demand charges constitute a substantial component of High Tension electricity bills in India. The present work incorporates this insight by explicitly evaluating measures according to their effect on recorded peak demand as well as on energy consumed.

2.2 COMPARISON AND DISCUSSION

The reviewed literature establishes strong methodological foundations for the individual components of the proposed system. Gradient boosting is well validated for building consumption regression; residual-based detection is established practice; non-intrusive load monitoring provides the conceptual basis for attribution; and additive explanation methods offer principled interpretability.

Three gaps emerge, however, when this body of work is considered against the requirements of an operational decision support system for an Indian campus block. First, attribution of anomalies to physical equipment is generally either absent or dependent on distinguishable appliance signatures that do not exist in an environment of hundreds of identical devices. Second, the academic calendar is rarely treated as a first-class modelling input despite exerting substantial influence on campus consumption; examination periods and vacations produce day types that generic weekday and weekend indicators cannot represent. Third, and most consequentially, published evaluations

optimise almost exclusively for energy consumed, whereas Indian High Tension consumers are billed on both energy and maximum demand, with the latter frequently forming the larger component.

CampusWatt addresses these three gaps directly. Attribution is achieved through inventory-constrained arithmetic rather than signature discrimination. The academic calendar is encoded explicitly as a set of day-type features, an addition that reduced forecast error in the computer laboratory zone from approximately twenty-seven percent to under three percent. Optimisation targets the complete tariff, which yields recommendations that a consumption-only formulation cannot reach, most notably the staggering of laboratory sessions to reduce coincident peak demand without altering total consumption.

2.3 CONCLUSION

The literature survey confirms that the constituent techniques required for intelligent campus energy management are individually mature, and that the contribution available lies not in the invention of new algorithms but in their integration into a system that produces decisions rather than measurements.

The survey further identifies the specific deficiencies the present work addresses: the absence of equipment-level attribution in environments containing homogeneous loads, the neglect of academic calendar structure in campus consumption modelling, and the failure to optimise against the maximum demand component of the applicable tariff. CampusWatt is designed to close these gaps within the scope of a single academic block, while remaining extensible to campus scale.

CHAPTER 3

SYSTEM DESIGN

This chapter presents the design of the proposed system. The overall architecture is described first, followed by the software, hardware, dataset and deployment requirements, and concluding with process models depicting system behaviour and data movement.

3.1 SYSTEM ARCHITECTURE

The architecture is organised into six layers, each consuming the output of the layer preceding it. This arrangement isolates concerns, permitting any single layer to be modified or replaced without disturbing the remainder of the system.

The Data Acquisition layer is responsible for producing the hourly consumption record. In the present implementation, consumption is generated from a parameterised physical model in which each zone's load at a given hour is computed as the sum over installed equipment of rated wattage multiplied by an occupancy-dependent and weather-dependent duty cycle. In deployment, this layer is replaced by ingestion from installed metering hardware; the interface presented to subsequent layers is identical in either case, which permits the analytical stack to be developed and validated in advance of hardware installation.

The Preprocessing and Feature Engineering layer constructs the model input matrix. Lag features are computed at one, twenty-four and one hundred and sixty-eight hour offsets, corresponding respectively to short-term persistence, daily periodicity and weekly periodicity. Rolling means and standard deviations summarise recent behaviour. Hour of day and day of week are encoded cyclically using sine and cosine transforms so that the model perceives hour twenty-three and hour zero as adjacent. Weather variables and their interactions with occupancy indicators are included, together with binary flags for each academic day type.

The Forecasting layer maintains one trained model per zone. Zone-level rather than block-level modelling is adopted because the zones exhibit fundamentally different behaviour: the server room presents an almost constant load, the computer laboratories a sharply session-driven profile, and the classrooms a temperature-sensitive one. A single aggregate model would be required to represent all three simultaneously and would consequently represent none of them well.

The Detection and Attribution layer operates on forecast residuals. Where the residual exceeds the configured significance threshold, the excess power is passed to the signature matching procedure. This procedure divides the excess by the rated wattage of each equipment category installed in the affected zone and compares the resulting quantity against the installed count, eliminating any explanation that would require more units than physically exist. The surviving explanations constitute a shortlist for maintenance investigation.

The Financial and Decision Support layer applies the tariff model to convert physical quantities into monetary terms, and hosts the explainability, simulation and recommendation modules. The Presentation layer exposes the entire system through a REST application programming interface consumed by a role-based single-page web application, with authentication by JSON Web Token and authorisation enforced at the endpoint level.

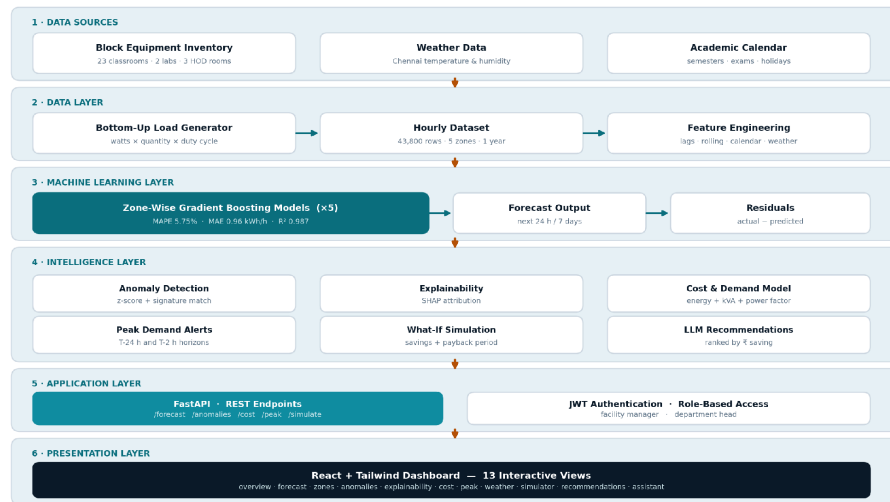


Figure 3.1 Architecture diagram of the proposed system

3.2 SYSTEM REQUIREMENTS

3.2.1 Software Requirements

The system is implemented in Python for all data processing, machine learning and service components, and in JavaScript for the user interface. The principal software dependencies are summarised in Table 3.1.

Table 3.1 Software requirements of the proposed system

Software Component	Version	Purpose
Python	3.12	Primary language for the backend, machine learning pipeline and data processing.
FastAPI	0.115	Provides the REST application programming interface and generates OpenAPI documentation.
Uvicorn	0.34	Asynchronous server gateway interface server hosting the backend application.
pandas	2.2	Time series manipulation, aggregation and feature construction.
NumPy	2.1	Numerical computation underlying the feature engineering routines.
scikit-learn	1.6	Histogram-based gradient boosting regression and evaluation metrics.
SHAP	0.52	Additive feature attribution for the explainability module.
joblib	1.4	Serialisation and loading of the five trained zone models.
SQLAlchemy	2.0	Object-relational mapping for users, acknowledgements and audit records.
PostgreSQL	16	Relational store for application state in deployment; SQLite in development.
React	18	Component-based user interface library for the web application.
Tailwind CSS	3.4	Utility-first styling framework for the interface.
bcrypt	4.2	Password hashing with per-credential salt generation.
python-jose	3.3	JSON Web Token encoding and verification.
pytest	8.3	Automated test framework for unit and integration testing.
Docker	Latest stable	Containerisation for reproducible deployment.

3.2.2 Hardware Requirements

The computational requirements of the system are modest, since model training operates on a dataset of approximately forty-four thousand records and inference is performed against pre-trained artefacts. The requirements are summarised in Table 3.2.

Table 3.2 Hardware requirements of the proposed system

Component	Specification	Purpose
Processor	Intel Core i5 or AMD Ryzen 5, four cores or above	Executes the backend service, database, model training and inference.
Memory	8 GB minimum, 16 GB recommended	The complete dataset and five models occupy approximately 250 MB when resident.
Graphics	Not required	Gradient boosting on tabular data of this scale does not benefit from acceleration.

Component	Specification	Purpose
Storage	256 GB solid state drive	Holds source code, dataset, trained models and the application database.
Metering	Three-phase energy meter with communication interface	Required for deployment against live data; not required for development.
Network	Stable broadband connection	Retrieval of weather forecasts and remote access to the interface.

3.2.3 Dataset Requirements

The system requires four categories of data. The consumption dataset comprises hourly kilowatt-hour readings for each functional zone across a full calendar year, yielding 43,800 records for five zones. Each record carries the timestamp, zone identifier, consumption, prevailing temperature and humidity, and the applicable academic day type.

The equipment inventory records the rated wattage and installed quantity of every device category within each zone. This inventory is not merely descriptive; it is functionally required by the attribution procedure, which cannot operate without knowledge of what is installed. The audited inventory for the block under study comprises twenty-three classrooms, two computer laboratories containing two hundred and forty desktop computers, three departmental head offices, three corridors, two washroom areas and one server room.

The weather record provides hourly temperature and relative humidity for the location, exhibiting realistic seasonal and diurnal variation for the Chennai climate, including the reduced temperatures associated with the north-east monsoon period.

The academic calendar assigns each day of the year to one of seven day types: working day, working Saturday, Sunday, examination period, vacation, declared holiday and institutional event day. This assignment constitutes a first-class model input rather than metadata.

An anomaly ground truth record accompanies the consumption dataset, documenting eleven injected events spanning six fault categories together with their affected zone, duration and magnitude. This record enables detection performance to be evaluated quantitatively, which is not possible with unlabelled public datasets.

3.2.4 Deployment and Scaling

The system is packaged for deployment as a containerised application. The backend service, together with its dependencies and trained model artefacts, is built into a Docker image using a multi-stage build in which compilation tooling is confined to the build stage and excluded from the runtime image. The container executes as an unprivileged user.

Application state is held in PostgreSQL, configured with connection pooling and pre-ping validation to prevent the delivery of connections that the database server has already closed. The consumption time series is loaded into memory once at service start, since it is read-only and re-reading it per request would impose unnecessary latency.

Two health endpoints are exposed. A liveness endpoint confirms that the process is responsive, while a readiness endpoint additionally verifies that the dataset and model artefacts have loaded successfully, permitting an orchestrator to withhold traffic from an instance that is running but not yet able to serve correct results. Structured logging with per-request correlation identifiers permits a single request to be traced across all layers.

For the scale under consideration, a single container instance behind a reverse proxy terminating Transport Layer Security is sufficient. Horizontal scaling and distributed caching are documented as future considerations but are not warranted by the present workload.

3.3 SYSTEM DESIGN

3.3.1 Activity Diagram

The activity diagram for the operational environment describes the sequence of actions performed by the system in processing consumption data and producing decisions. The sequence begins with the arrival of a meter reading and proceeds through feature construction, forecast generation, residual computation and threshold evaluation.

Where the residual is not significant, the reading is recorded and the sequence terminates. Where the residual is significant, control passes to the signature matching procedure, which produces a shortlist of candidate equipment explanations. The financial layer then computes the monetary impact of the event, after which an alert is raised to the facility manager. The manager may acknowledge the finding, whereupon an audit record is written, or may allow it to remain open.

The activity diagram for the maintenance environment describes the periodic retraining sequence. On the scheduled interval, newly accumulated consumption data is validated, the feature pipeline is re-executed, and each zone model is retrained. The retrained models are evaluated against the retained test partition, and are promoted to serving only where their error metrics do not exceed those of the incumbent models. Where degradation is detected, the incumbent models are retained and an alert is raised for investigation.

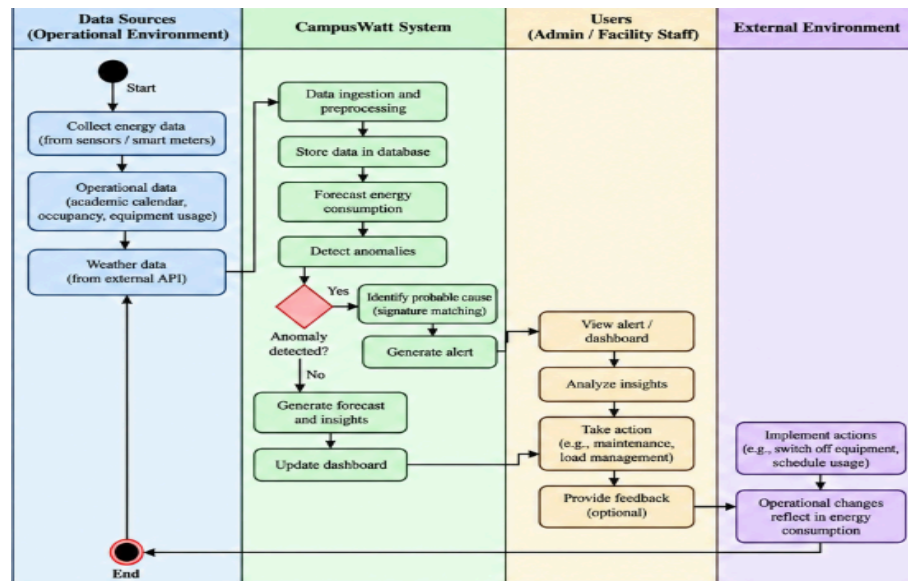


Figure 3.2 Activity diagram for the operational environment

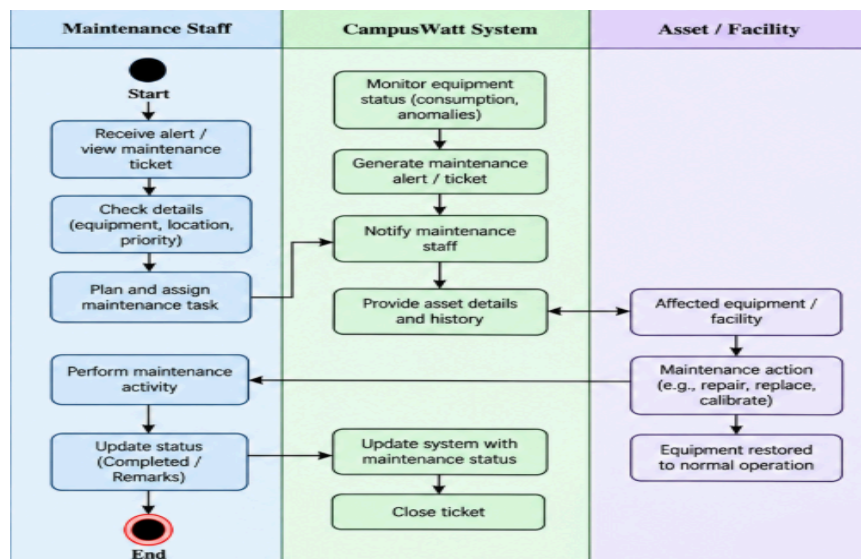


Figure 3.3 Activity diagram for the maintenance environment

3.3.2 Data Flow Diagram

The data flow diagram represents the movement of data through the system independently of control flow. Meter readings enter the Ingestion process, which normalises them and writes them to the Consumption Store. The Feature Engineering process reads from this store together with the Weather Store and the Academic Calendar Store, producing a feature matrix consumed by the Forecasting process.

The Forecasting process reads the trained model artefacts from the Model Store and emits predictions, which are compared against measured values by the Residual Analysis process. Significant residuals flow to the Attribution process, which reads the Equipment Inventory Store and emits attributed findings to the Findings Store.

The Tariff process reads consumption and demand data and applies the configured tariff parameters, emitting monetary values consumed by the Recommendation process. The Recommendation process additionally reads from the Findings Store and emits ranked measures to the presentation interface. Acknowledgements and configuration changes entered by the facility manager flow back into the Application State Store, and the Audit process records every state-changing action.

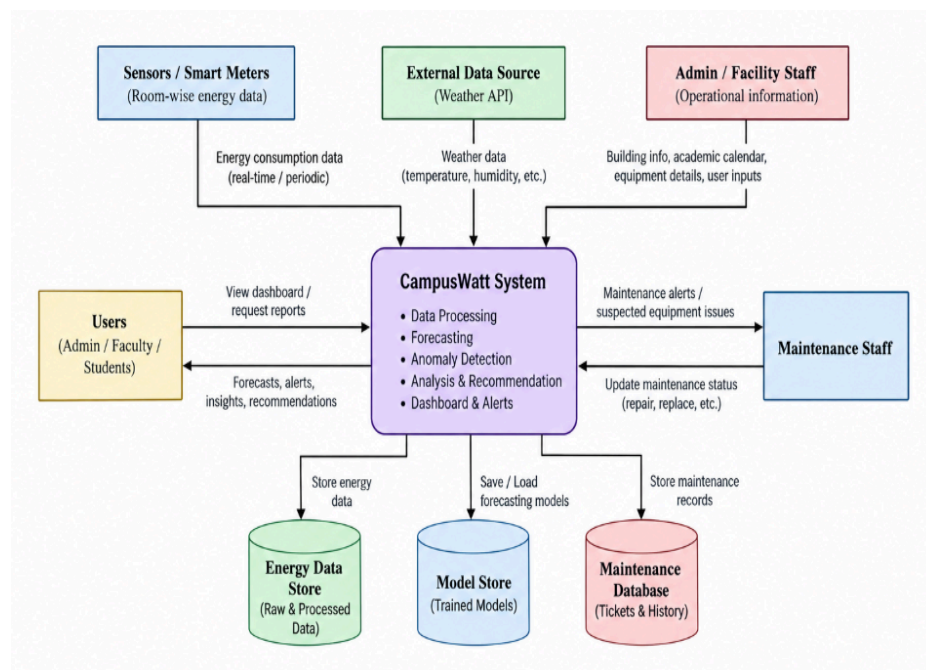


Figure 3.4 Data flow diagram of the system

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

The methodology adopted proceeds through six sequential stages. Consumption data is first constructed from the audited equipment inventory combined with weather and calendar inputs. Features are then engineered to expose temporal, meteorological and scheduling structure to the learning algorithm. A gradient-boosting regression model is trained for each zone and evaluated against a held-out temporal partition, using a seasonal-naive predictor as the comparison baseline.

Forecast residuals are subsequently analysed to identify anomalous consumption, and detected excess is attributed to equipment categories through inventory-constrained arithmetic. The complete tariff is then applied to express all findings in monetary terms. Finally, candidate efficiency measures are evaluated for annual saving, capital requirement, payback period and effect on peak demand, and are ranked for presentation.

A principle observed throughout is that no figure presented to the user is generated by inference where it can be retrieved by computation. The conversational assistant, in particular, is constrained to phrase values retrieved from the analytical layers and is prohibited from performing arithmetic of its own, since a model permitted to compute its own figures will eventually produce one that is incorrect.

4.2 MODULES

4.2.1 Dataset Generation and Preprocessing

The dataset is generated from a parameterised physical model rather than obtained from a public repository. The rationale is twofold. First, no public dataset carries the equipment inventory that the attribution procedure requires. Second, public consumption datasets are unlabelled with respect to anomalies, which prevents quantitative evaluation of detection performance.

Consumption for each zone at each hour is computed as the sum across equipment categories of rated wattage multiplied by installed quantity multiplied by a duty cycle. The duty cycle is itself a function of the occupancy factor for the hour and day type, and, for thermally driven loads such as fans and air conditioning, of the prevailing outside temperature. Lighting duty additionally responds to humidity as a proxy for overcast

conditions. Metering and behavioural noise is introduced as a multiplicative Gaussian term.

The weather model reproduces the Chennai climate, combining a seasonal sinusoid peaking in late May with a diurnal cycle peaking in mid-afternoon, adjusted downward during the north-east monsoon months. The academic calendar encodes semester boundaries, examination periods, vacations, declared holidays and institutional events.

Eleven anomalies spanning six categories are then injected at known locations and magnitudes: overnight load, unscheduled weekend operation, consumption during declared holidays, gradual baseline creep representing equipment degradation, acute equipment fault, and meter under-reporting. The injected events and their parameters are recorded separately as ground truth. The resulting dataset comprises 43,800 hourly records across five zones for one calendar year, totalling 147,155 kilowatt-hours.

Table 4.1 Audited equipment inventory of the academic block

Zone	Rooms	Fans	Lights	Computers	Panels	AC	Serve rs
Classrooms	23	224	135	—	20	—	—
Computer Labs	2	30	12	240	2	—	—
HOD Rooms	3	3	9	6	—	3	—
Common Areas	5	—	70	—	—	—	—
Server Room	1	—	2	—	—	1	2

4.2.2 Feature Engineering

Feature engineering constructs the matrix presented to the learning algorithm. Lag features capture temporal dependence at one hour, twenty-four hours and one hundred and sixty-eight hours, representing immediate persistence, daily periodicity and weekly periodicity respectively. Rolling means and standard deviations over twenty-four and one hundred and sixty-eight hour windows summarise recent level and volatility, and are shifted to exclude the current observation and thereby prevent leakage.

Hour of day and day of week are encoded cyclically through sine and cosine transformation, ensuring that the model treats adjacent hours as adjacent across the midnight boundary. Weather features comprise temperature, relative humidity, a derived

cooling degree quantity computed as the excess of temperature above twenty-five degrees Celsius, and an interaction term between temperature and the working-day indicator.

The most influential single feature is a learned profile of mean consumption for each combination of hour and academic day type, computed exclusively from the training partition to prevent leakage. This feature supplies the model with an explicit representation of what a given hour of a given day type ordinarily consumes, which proves decisive during regime transitions when all lagged values originate from a different day type. Its introduction reduced computer laboratory forecast error from approximately twenty-seven percent to under three percent.

Additional features indicate whether the day type of the observation matches that of its lagged counterparts, permitting the model to discount lag features across regime boundaries.

Table 4.2 Feature set used for model training

Feature Group	Features	Purpose
Lag	lag 1 h, 24 h, 48 h, 168 h	Short-term persistence, daily and weekly periodicity
Rolling	mean and standard deviation over 24 h and 168 h	Recent level and volatility, shifted to prevent leakage
Cyclical	sine and cosine of hour and weekday	Continuity across the midnight and week boundaries
Weather	temperature, humidity, cooling degrees	Thermally driven fan and cooling load
Calendar	working day, exam period, vacation, event	Academic day-type behaviour
Profile	mean consumption by hour and day type	Expected value during regime transitions

4.2.3 Model Training and Evaluation

A histogram-based gradient boosting regressor is trained independently for each of the five zones. Independent models are preferred to a single multi-zone model because the zones differ fundamentally in dynamics; the server room exhibits an almost constant profile while the computer laboratories are sharply session-driven, and a single model would compromise between these behaviours rather than represent either accurately.

The dataset is partitioned temporally rather than randomly, with January to September forming the training partition and October to December the test partition.

Temporal partitioning is essential for time series evaluation, since random partitioning would permit the model to observe future values when predicting the past, producing optimistic and invalid accuracy estimates.

Training is performed exclusively on records not marked as anomalous. This ensures that the model learns normal operating behaviour, so that anomalous periods produce large residuals and thereby become detectable. Training on contaminated data would cause the model to partially learn the anomalies and consequently fail to flag them.

Evaluation is conducted against a seasonal-naive baseline that predicts consumption equal to that observed at the same hour one week previously. Reporting improvement against a stated baseline is necessary because an absolute error figure conveys little without reference to the difficulty of the prediction task.

Table 4.3 Model configuration and training parameters

Parameter	Value
Algorithm	Histogram-based gradient boosting regression
Models trained	Five, one per functional zone
Training partition	January to September (temporal split)
Test partition	October to December, unseen during training
Maximum iterations	700 with early stopping
Learning rate	0.05
Maximum depth	7
L2 regularisation	1.0
Validation fraction	0.12
Baseline for comparison	Seasonal naive at 168-hour lag

4.2.4 Anomaly Detection and Signature Matching

Anomaly detection operates on forecast residuals rather than on raw consumption comparisons. Comparison against the previous day would generate false alarms at every regime transition, flagging each Monday following a Sunday and each working day following a holiday. Comparison against the forecast is conditioned on weather, day type and schedule, and therefore isolates genuinely unexpected behaviour.

Two detectors operate in parallel. The residual detector identifies acute events by flagging observations whose standardised residual exceeds the configured threshold. The

rolling baseline monitor identifies gradual degradation by tracking week-on-week drift in the zone baseline, which is invisible to the residual detector because each individual hour departs only slightly from expectation.

Attribution is performed by signature matching. Given an excess power quantity and the affected zone, the procedure computes for each installed equipment category the number of units that would be required to account for the excess, obtained by dividing the excess power by the rated wattage of that category. Where the required quantity exceeds the installed quantity, the explanation is physically impossible and is eliminated. The surviving candidates constitute the shortlist reported to the operator.

As an illustration, an overnight excess of nineteen kilowatts in the computer laboratory zone corresponds to approximately one hundred and twenty-seven desktop computers at one hundred and fifty watts each, which is feasible given the two hundred and forty installed. The same excess would require two hundred and fifty-three ceiling fans, of which only thirty are installed, and is therefore eliminated. The system consequently reports that the event is consistent with computers left powered rather than with fans left running.

It is acknowledged that with a single block-level meter this procedure identifies a probable cause rather than a confirmed one. Sub-metering at floor level would permit definitive attribution and is identified as future work.

4.2.5 Tariff Modelling and Explainability

The tariff module implements the complete High Tension commercial tariff structure applicable to institutional consumers. The energy charge is computed with time-of-day banding, applying a surcharge during the evening peak window and a rebate during the night window. The maximum demand charge is computed on the highest demand recorded during the billing period, irrespective of the duration for which that demand persisted. The power factor penalty is applied proportionally where the measured power factor falls below the prescribed target. Statutory electricity duty and fixed charges complete the computation.

The significance of modelling the demand charge separately cannot be overstated. Analysis of the block under study establishes that the demand charge component exceeds the energy charge component. A system optimising solely for kilowatt-hours consumed

would therefore address the smaller portion of the bill while neglecting the larger. It is this observation that generates the highest-value recommendation produced by the system, namely the staggering of laboratory sessions to reduce coincident peak demand while leaving total consumption unchanged.

The explainability module decomposes period-on-period consumption change into contributing factors. The weather contribution is computed from the change in cooling degree days multiplied by the measured zone sensitivity. The calendar contribution derives from the change in the count of working days. The anomaly contribution is obtained from detected events within the period. The residual, being the portion not accounted for by these factors, is reported explicitly rather than distributed among them, on the principle that an honest statement of unexplained variation is more useful than a forced attribution.

4.2.6 Simulation, Recommendation and Presentation

The what-if simulator permits arbitrary combinations of candidate efficiency measures to be evaluated. For each selected combination the system computes the aggregate annual saving, the total capital requirement, the blended payback period and the resulting projected maximum demand. Where a selected combination reduces projected demand below the sanctioned limit, this is reported explicitly, since crossing that threshold eliminates the demand penalty entirely and therefore represents a discontinuity in benefit rather than an incremental improvement.

The recommendation engine ranks measures by annual monetary saving and tiers them by capital requirement into zero-cost, low-cost and capital categories. Each recommendation carries the evidence that generated it, a nominated owner, the estimated saving, the capital requirement, the payback period and a sequence of concrete implementation steps.

Ranking by monetary saving rather than by technical appeal produces an ordering that is occasionally counter-intuitive. The replacement of legacy fluorescent fittings with light-emitting diode equivalents, a measure commonly proposed first in institutional energy discussions, ranks last in the generated list because its annual saving is small relative to the behavioural and scheduling measures above it. This ordering is retained deliberately, as it demonstrates that the engine prioritises measured impact over familiarity.

The presentation layer delivers these outputs through a role-based web interface distinguishing a facility manager role, which may acknowledge findings and record actions, from a departmental head role restricted to read access. A conversational assistant accepts natural language queries, classifies them across a defined intent taxonomy, resolves contextual references to preceding turns, and responds using values retrieved from the analytical layers. Queries falling outside the available data are declined rather than answered speculatively.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

This chapter presents the implementation of the proposed system and the results obtained. The dataset characteristics are described first, followed by forecasting accuracy, detection and attribution outcomes, financial analysis, and the resulting recommendations.

5.1 IMPLEMENTATION

The system is implemented as a two-tier application. The backend is developed in Python using the FastAPI framework, exposing twenty REST endpoints covering monitoring, detection, financial analysis, decision support and the conversational assistant. Authentication is implemented using JSON Web Tokens with bcrypt password hashing, and authorisation is enforced at endpoint level with two roles.

The frontend is a single-page application implemented in React with Tailwind CSS, comprising thirteen views. Interface data is obtained exclusively through a single abstraction module, which permits the application to operate against either the live backend or a local fixture set without modification to any view.

Correctness is verified by an automated test suite of 297 tests achieving 92 percent statement coverage. The suite comprises unit tests for the tariff arithmetic and attribution logic, integration tests exercising every endpoint through the complete application stack, and tests confirming that role restrictions are enforced. Two tests assert the central financial finding directly, failing should the relationship between demand and energy charges be inverted by a configuration error.

5.2 DATASET CHARACTERISTICS

The generated dataset comprises 43,800 hourly records across five zones for one calendar year, representing 147,155 kilowatt-hours of consumption. Validation against expected physical behaviour confirms a night baseline of 4.5 kilowatts during unoccupied hours, a peak demand of 162.7 kilowatts, a visible reduction during the lunch interval, and an annual maximum during April consistent with the Chennai summer.

Zone-wise distribution of annual consumption is presented in Figure 5.1. The computer laboratories account for 48.7 percent of block consumption, followed by classrooms at 29.1 percent, the server room at 10.0 percent, common areas at 6.4 percent

and departmental head offices at 5.7 percent. This distribution is itself an analytically significant result, since it establishes that any efficiency measure not addressing the laboratories operates upon the minority of consumption.

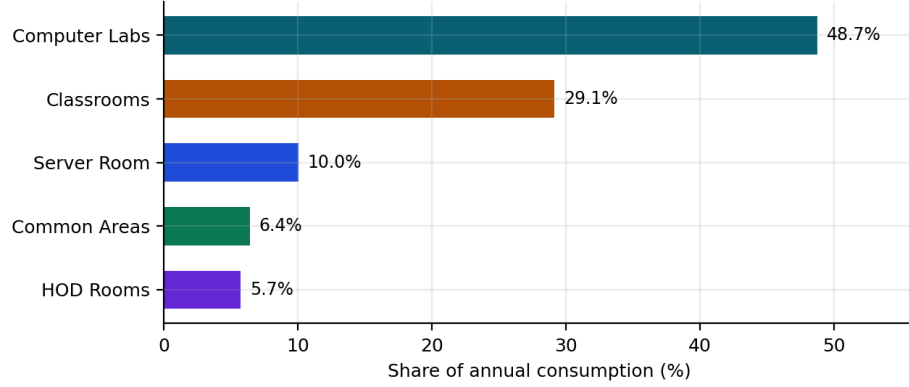


Figure 5.1 Zone-wise share of block energy consumption

5.3 FORECASTING RESULTS

Forecast accuracy is evaluated on the October to December partition, which the models did not observe during training. At block level the system attains a Mean Absolute Percentage Error of 5.75 percent, a Mean Absolute Error of 0.96 kilowatt-hours per hour and a coefficient of determination of 0.987. Figure 5.2 presents measured against forecast consumption for a representative working day.

Zone-level results are presented in Table 5.1 together with the corresponding seasonal-naive baseline. Improvement over baseline ranges from 51 percent in the server room, where the baseline is already strong owing to the near-constant load, to 98 percent in the computer laboratories, where the baseline performs poorly because session scheduling varies between weeks. Figure 5.3 presents this comparison graphically.

Two zones report elevated percentage error. The classrooms record 22.9 percent and the departmental head offices 25.8 percent. These figures are reported without adjustment, together with the corresponding Mean Absolute Errors of 0.73 and 0.10 kilowatt-hours per hour respectively. The discrepancy arises because Mean Absolute Percentage Error divides by the measured value, so a zone with small absolute consumption reports a large percentage for a small absolute error. The departmental head offices constitute 5.7 percent of block consumption, and their error consequently exerts negligible influence on the block-level figure. Classroom error is further concentrated in the transition hours of seven and seventeen, during which fan and lighting switching is

governed by occupant behaviour rather than by thermal conditions and is therefore inherently less predictable.

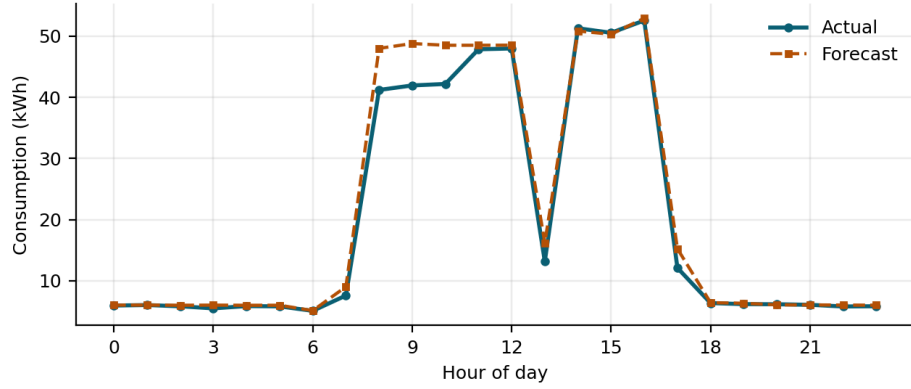


Figure 5.2 Block-level actual versus forecast consumption

Table 5.1 Zone-wise forecast accuracy against baseline

Zone	Share (%)	MAPE (%)	MAE (kWh/h)	Baseline MAPE (%)	Improve ment (%)
Computer Labs	48.7	5.6	0.250	142.10	98.3
Classrooms	29.1	22.9	0.733	106.94	80.8
Server Room	10.0	1.6	0.023	3.34	51.2
Common Areas	6.4	4.3	0.047	37.67	89.3
HOD Rooms	5.7	25.8	0.102	29.21	19.0
Block level	100.0	5.75	0.960	—	—

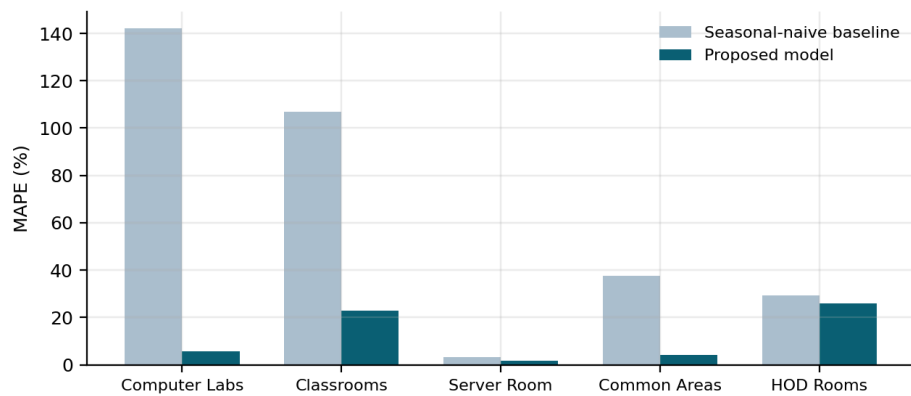


Figure 5.3 Forecast error against seasonal-naïve baseline, by zone

5.4 ANOMALY DETECTION AND ATTRIBUTION

The detection subsystem identifies eleven anomalous events across six fault categories, summarised in Table 5.2. Because the injected events are documented as ground truth, detection outcomes can be verified rather than merely asserted.

The most financially significant single finding is a sustained seven kilowatt draw persisting across four days of vacation in the computer laboratory zone, with no timetabled session. Signature matching attributes this to approximately forty-seven desktop computers remaining powered, an explanation consistent with the two hundred and forty installed. The alternative explanation of ceiling fans is eliminated, as it would require ninety-three units against thirty installed. The absence of any lighting component further indicates that the rooms were not occupied.

A second finding of particular interest is a gradual increase of approximately seventy-five percent in the departmental head office baseline over a three-week period, unaccompanied by any corresponding change in weather or occupancy. This pattern is characteristic of a degrading air conditioning compressor. Such degradation is invisible to day-to-day comparison, and its detection demonstrates the value of the rolling baseline monitor operating alongside residual analysis.

A meter under-reporting event is deliberately reported in a separate category rather than as a saving. Consumption cannot fall to near zero while rooms are timetabled and occupied, and treating such a reading as a genuine reduction would corrupt the baseline for subsequent periods. Detecting under-consumption as well as over-consumption addresses a failure mode that threshold-based systems typically overlook.

Table 5.2 Anomalies detected with attribution and estimated cost

Type	Zone	Hours	Cost (Rs)	Attribution
Vacation load	Computer Labs	52	2,940	≈ 47 computers powered during vacation
Overnight load	Computer Labs	12	1,835	≈ 127 computers; fans eliminated
Overnight load	Computer Labs	12	1,780	Recurrence of the same pattern
Unscheduled weekend	Computer Labs	8	1,420	Computers and lighting both present
Unscheduled weekend	Computer Labs	8	1,390	Unlogged Sunday session

Type	Zone	Hours	Cost (Rs)	Attribution
Baseline creep	HOD Rooms	400	1,240	AC compressor drawing 26% above rated
Holiday consumption	Classrooms	10	980	≈ 140 fans; no display panels active
Equipment fault	Computer Labs	5	640	Abrupt onset and recovery
Equipment fault	Common Areas	5	210	Exceeds listed inventory capacity
Overnight load	HOD Rooms	12	190	Exact match to one AC unit
Meter under-reporting	Classrooms	8	—	96% below expected while timetabled

5.5 FINANCIAL ANALYSIS AND RECOMMENDATIONS

Financial analysis is conducted using the High Tension commercial tariff. The composition of the monthly bill is presented in Figure 5.4. The demand charge of Rs 94,366 exceeds the energy charge of Rs 43,214, confirming that maximum demand rather than energy consumption constitutes the larger portion of the bill.

This finding governs the recommendation ordering. Peak demand for the period reaches 162.7 kilovolt-amperes against a sanctioned 150, a breach of 12.7 kilovolt-amperes attracting a penalty applied to the whole billing period. Figure 5.5 presents monthly recorded peaks against the sanctioned limit. Coincidence analysis establishes that the computer laboratories contribute the majority of the coincident peak, both laboratories being in session simultaneously during the early afternoon.

Consequently the measure of offsetting one laboratory session by one hour is estimated to reduce peak demand by approximately twenty-four kilovolt-amperes, sufficient to return the block within its sanctioned limit. This measure alters no total consumption whatsoever and requires no capital expenditure, yet is estimated to save Rs 43,200 annually. A system optimising solely against energy consumed could not have identified it.

The complete set of recommendations is presented in Table 5.3, totalling Rs 2,37,264 in estimated annual savings against Rs 66,000 of capital expenditure. The two highest-ranked measures require no capital at all.

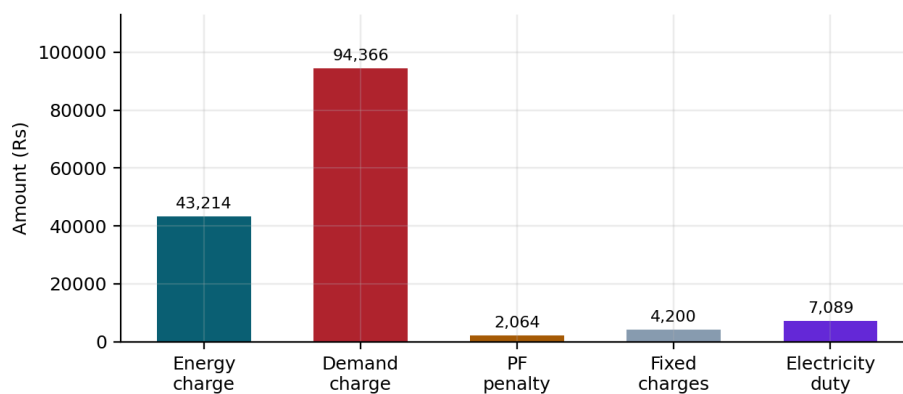


Figure 5.4 Composition of the monthly HT electricity bill

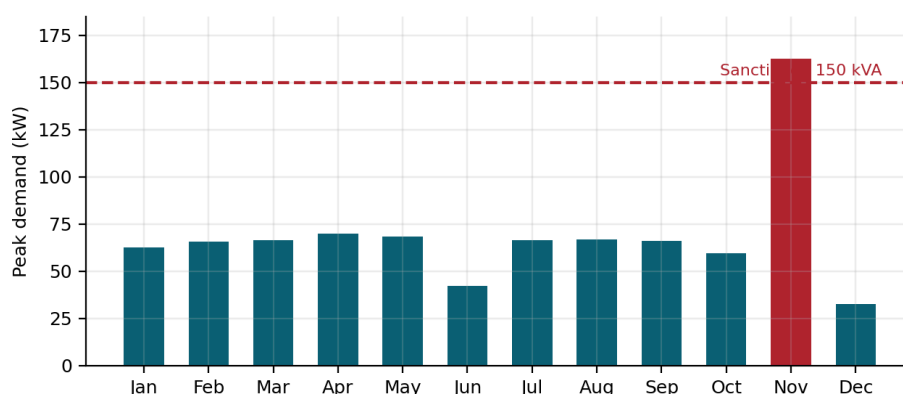


Figure 5.5 Monthly recorded peak demand against sanctioned limit

Table 5.3 Ranked savings recommendations

Ran k	Measure	Tier	Saving (Rs/yr)	Capital (Rs)	Payback
1	Sleep-on-idle across 240 laboratory computers	No cost	98,400	Nil	Immedia te
2	Offset one laboratory session by one hour	No cost	43,200	Nil	Immedia te
3	Fan discipline advisory on cool days	No cost	34,800	Nil	Immedia te
4	Timer control on corridor and washroom lighting	Low cost	29,600	18,000	7.3 months
5	Power factor correction capacitor bank	Capital	19,464	42,000	25.9 months
6	Air conditioning setpoint 24 °C to 26 °C	No cost	7,900	Nil	Immedia te
7	Replace 15 legacy fittings with LED	Capital	3,900	6,000	18.5 months

5.6 DISCUSSION

The results establish that the proposed system achieves its stated objectives within the scope of a single academic block. Forecast accuracy of 5.75 percent at block level, representing an improvement of between 80 and 98 percent over the seasonal-naïve baseline, is sufficient to render anomalous consumption statistically detectable.

The contribution of the academic calendar features warrants particular emphasis. Prior to their introduction, computer laboratory forecast error stood at approximately twenty-seven percent, the model being unable to distinguish an examination-period weekday from an ordinary teaching day when all lagged values originated from the latter. Introduction of the hour and day-type consumption profile as an explicit feature reduced this error to under three percent..

The signature matching procedure achieves its intended purpose of converting detection into diagnosis. Reporting that a zone drew nineteen kilowatts more than expected is of limited operational value; reporting that this is consistent with approximately one hundred and twenty-seven computers remaining powered, and inconsistent with any fan-based explanation, is directly actionable. The elimination criterion is arithmetically simple yet effective precisely because the equipment inventory is known. The financial analysis produced the least anticipated result of the work. The expectation prior to implementation was that reducing consumption would reduce cost proportionally. Modelling the complete tariff established instead that the demand charge exceeds the energy charge, and therefore that a brief coincident peak carries greater financial consequence than a sustained moderate load. This finding altered the recommendation ordering materially and is not obtainable from consumption data alone.

Certain limitations are acknowledged. The consumption dataset is synthesised rather than measured; while the underlying physical relationships are grounded in an audited inventory and the anomalies are labelled by construction, validation against installed metering is necessary before the accuracy figures can be regarded as characterising real deployment. Attribution identifies a probable rather than a confirmed cause, a limitation inherent to single-point metering. The tariff parameters are indicative and require substitution with figures from an actual bill. Finally, the evaluation covers one block over one year; generalisation to other buildings and longer horizons remains to be established.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This work presents CampusWatt, an integrated platform for energy forecasting and decision support in campus buildings, developed and evaluated for a single academic block. The system combines zone-level consumption forecasting, forecast-residual anomaly detection with equipment-level attribution, complete tariff modelling, explainability, scenario simulation and ranked recommendation within a single pipeline.

The forecasting component attains a block-level Mean Absolute Percentage Error of 5.75 percent on a held-out temporal partition, improving upon a seasonal-naïve baseline by between 80 and 98 percent across zones. The detection component identifies eleven anomalous events across six fault categories and attributes each to a shortlist of candidate equipment through inventory-constrained signature matching. The financial component models the complete High Tension tariff and establishes that the maximum demand charge exceeds the energy charge for the block under study.

The recommendation engine identifies Rs 2,37,264 in estimated annual savings, representing approximately twenty percent of the annual electricity cost, of which the two highest-ranked measures require no capital expenditure. The highest-value peak reduction measure, the offsetting of laboratory sessions by one hour, alters no total consumption and would not have been identified by a system optimising against energy alone.

The principal contribution of the work lies not in the forecasting algorithm, which employs established methods, but in the decision layer constructed above it: attribution of anomalies to physical equipment in an environment of homogeneous loads, explicit modelling of academic calendar structure, and optimisation against the complete applicable tariff rather than against consumption alone.

6.2 FUTURE WORK

Several directions for extension are identified. The most immediate is the installation of physical metering and validation of the models against measured consumption, which would replace the synthesised dataset and permit the reported accuracy figures to be confirmed under operational conditions.

Sub-metering at floor or circuit level would resolve the principal limitation of the attribution procedure. With a single block-level meter the system narrows the cause of an anomaly to a shortlist; with circuit-level measurement it could identify the affected circuit definitively, converting probable attribution into confirmed diagnosis.

Integration of live weather forecasting through a public meteorological service would extend the advisory function from retrospective analysis to genuine day-ahead planning, permitting fan and cooling load guidance to be issued before the day in question rather than after it.

Extension from a single block to the full campus would introduce inter-building comparison, allowing structurally similar blocks to serve as mutual controls. Divergence between two comparable buildings constitutes evidence of a fault in one of them without requiring any forecast at all.

Further extensions include the incorporation of solar photovoltaic generation and its effect on net demand, forecasting of the monthly maximum demand to permit pre-emptive load management, and automated scheduling recommendations that account jointly for academic timetable constraints and tariff structure.

REFERENCES

- [1] K. Amasyali and N. M. El-Gohary, "A review of data-driven building energy consumption prediction studies," *Renewable and Sustainable Energy Reviews*, vol. 81, pp. 1192–1205, 2018.
- [2] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 785–794.
- [3] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Advances in Neural Information Processing Systems* 30, 2017, pp. 3146–3154.
- [4] J. E. Seem, "Using intelligent data analysis to detect abnormal energy consumption in buildings," *Energy and Buildings*, vol. 39, no. 1, pp. 52–58, 2007.
- [5] Y. Himeur, K. Ghanem, A. Alsalemi, F. Bensaali, and A. Amira, "Artificial intelligence based anomaly detection of energy consumption in buildings: A review, current trends and new perspectives," *Applied Energy*, vol. 287, Art. no. 116601, 2021.
- [6] G. W. Hart, "Nonintrusive appliance load monitoring," *Proceedings of the IEEE*, vol. 80, no. 12, pp. 1870–1891, 1992.
- [7] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems* 30, 2017, pp. 4765–4774.
- [8] C. Miller et al., "The Building Data Genome Project 2, energy meter data from the ASHRAE Great Energy Predictor III competition," *Scientific Data*, vol. 7, Art. no. 368, 2020.
- [9] C. Fan, F. Xiao, and S. Wang, "Development of prediction models for next-day building energy consumption and peak power demand using data mining techniques," *Applied Energy*, vol. 127, pp. 1–10, 2014.
- [10] L. Pérez-Lombard, J. Ortiz, and C. Pout, "A review on buildings energy consumption information," *Energy and Buildings*, vol. 40, no. 3, pp. 394–398, 2008.